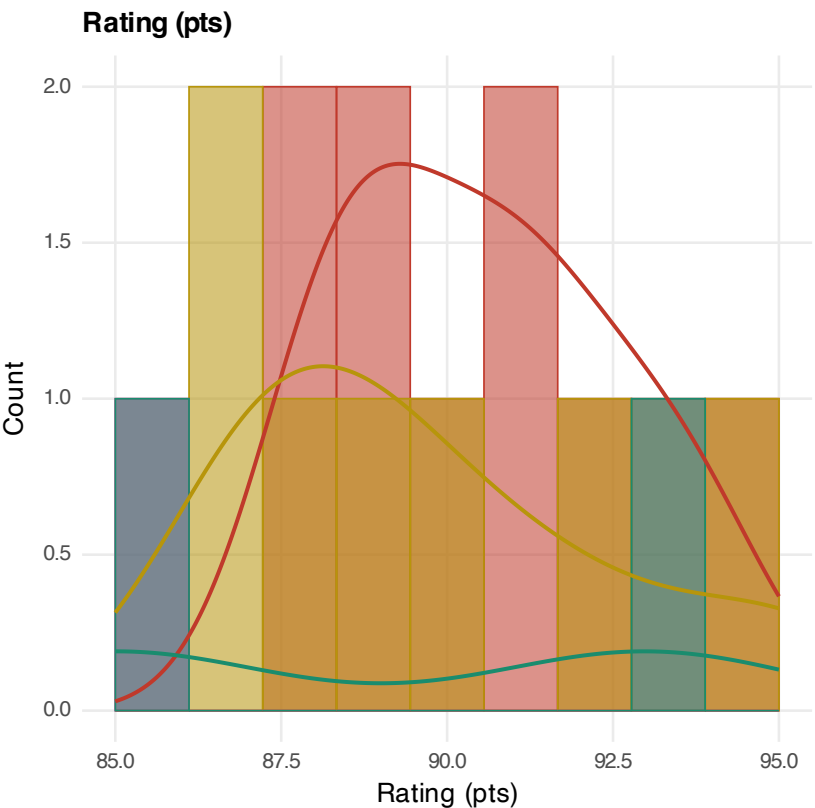
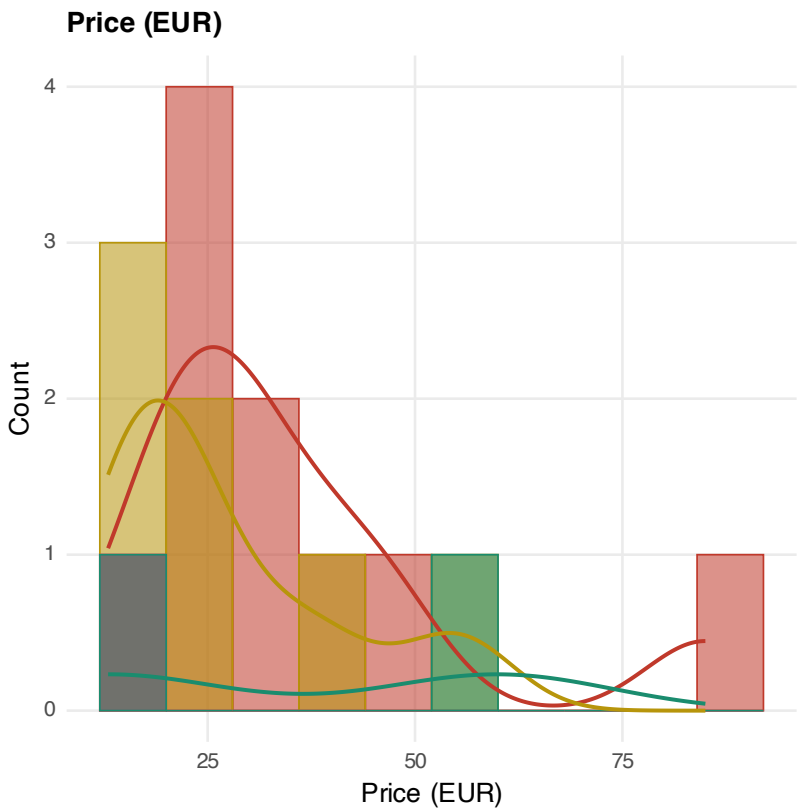
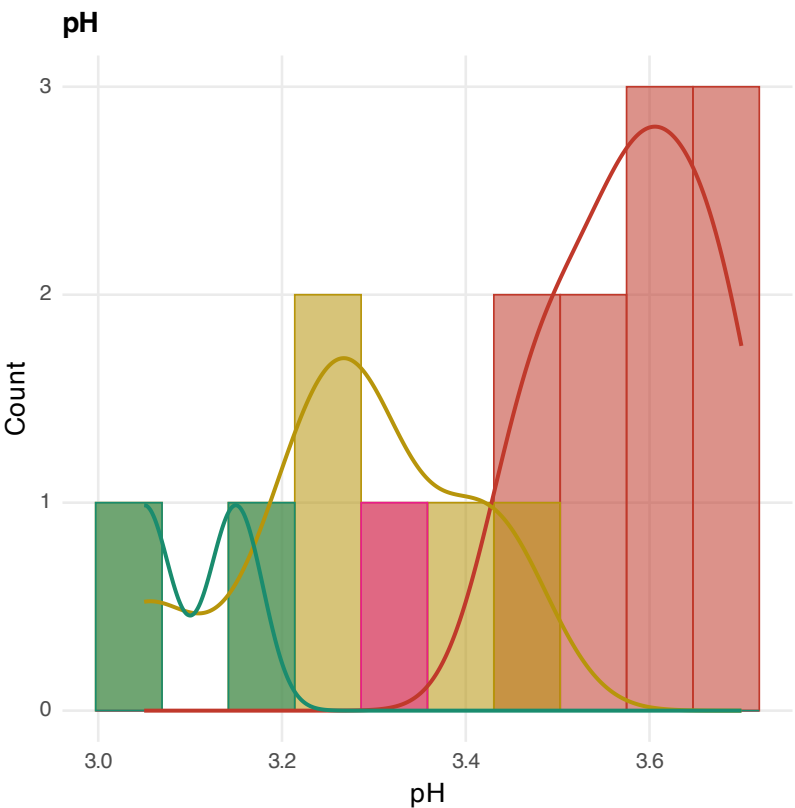
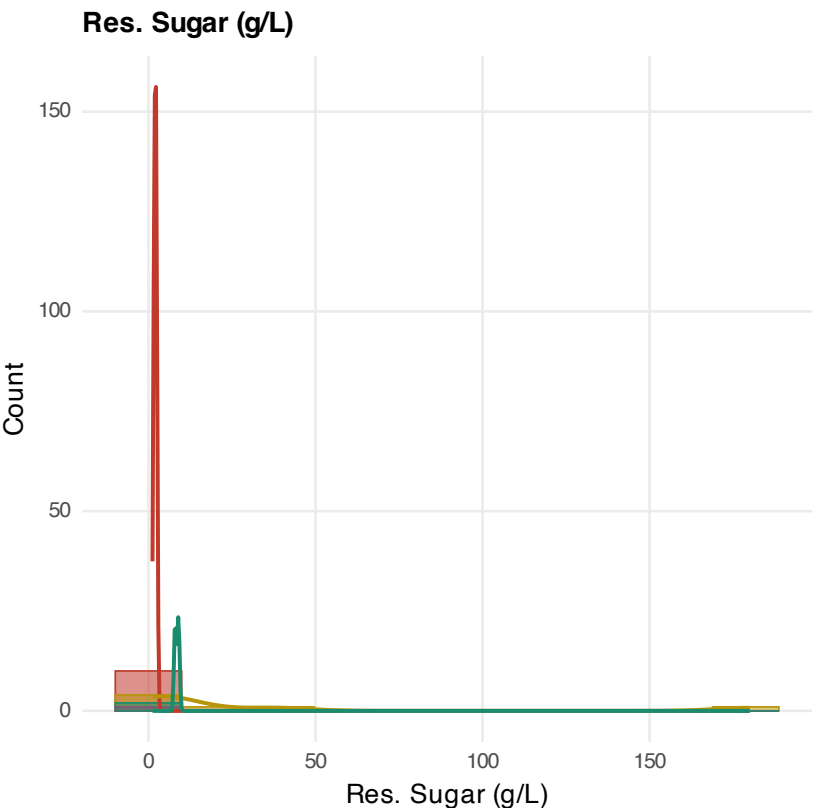
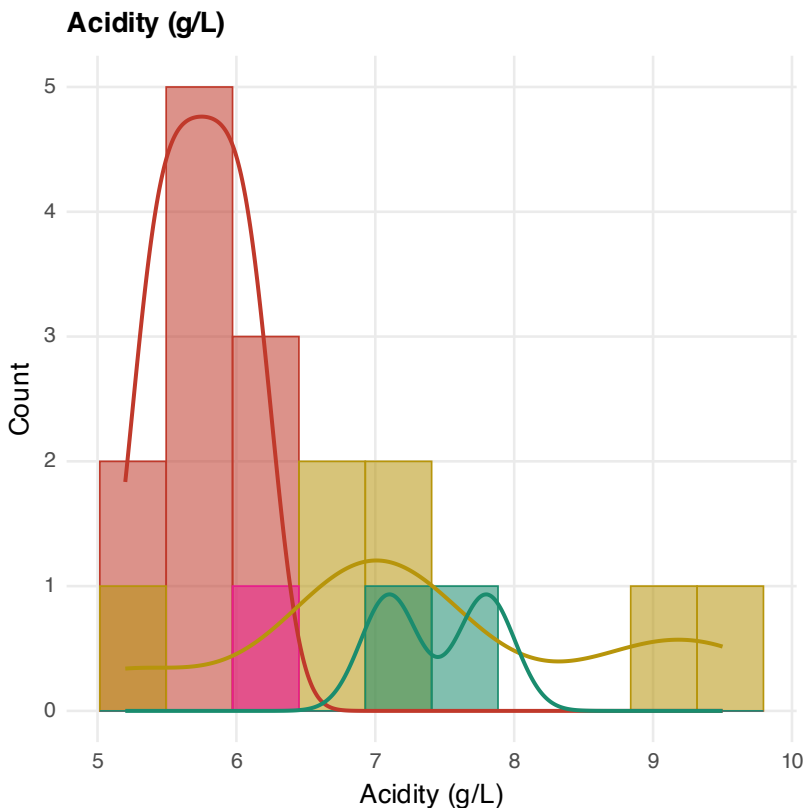
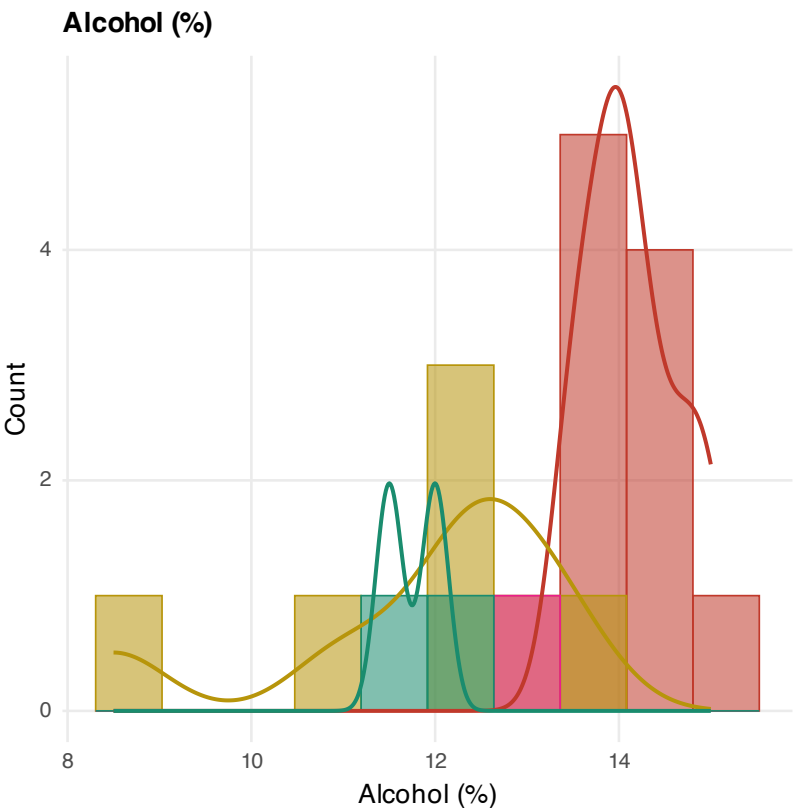
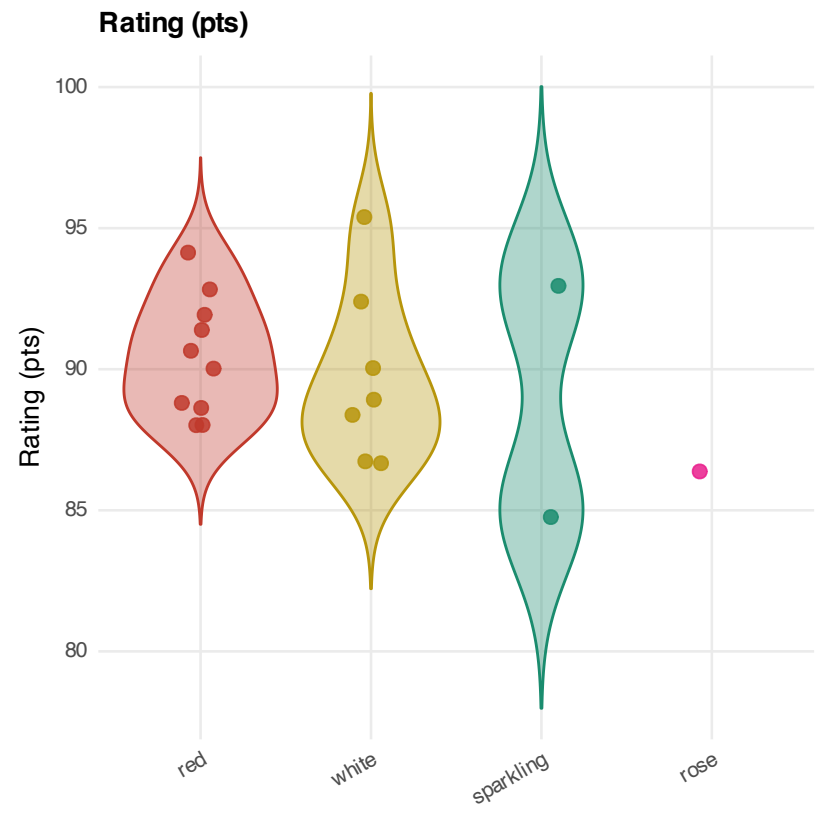
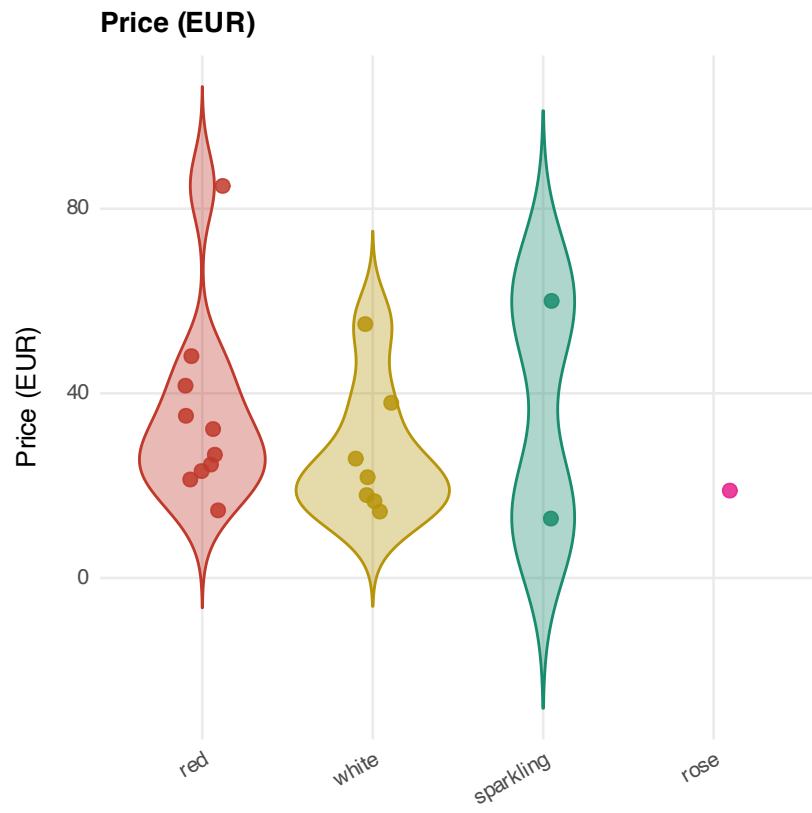
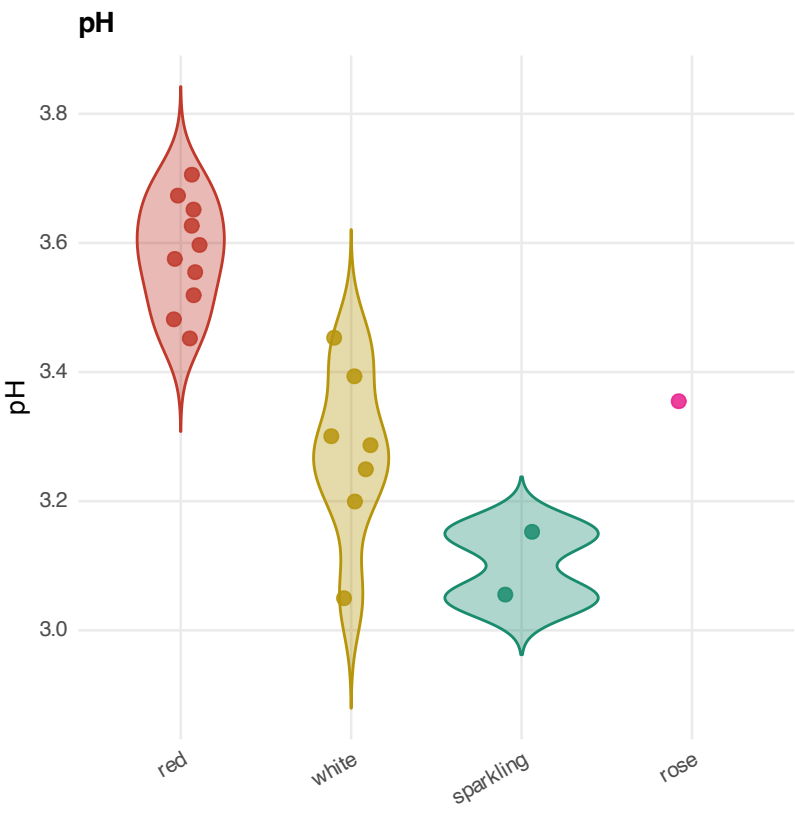
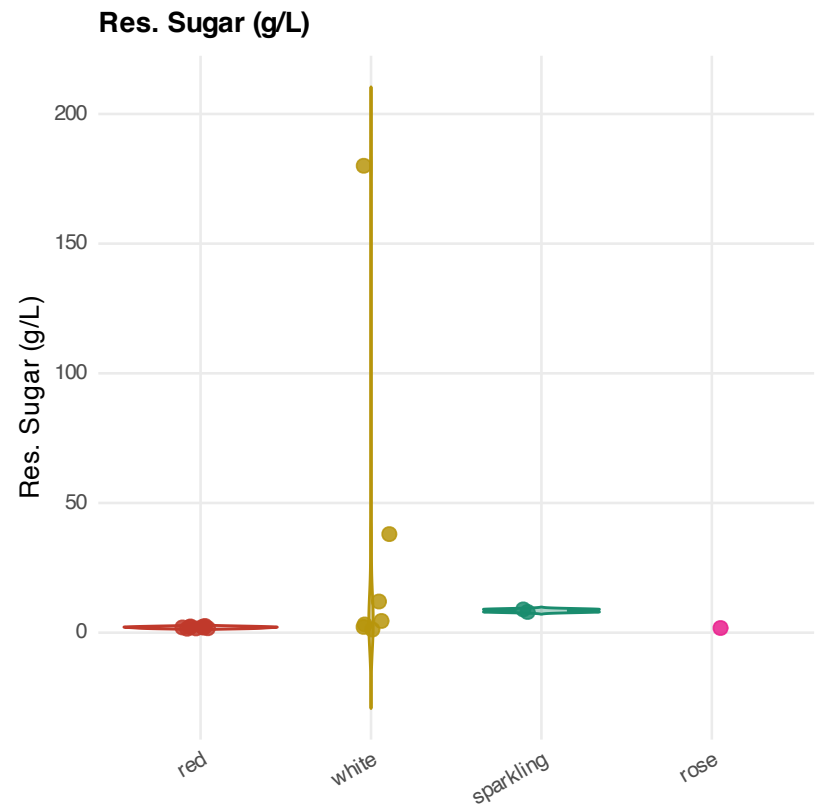
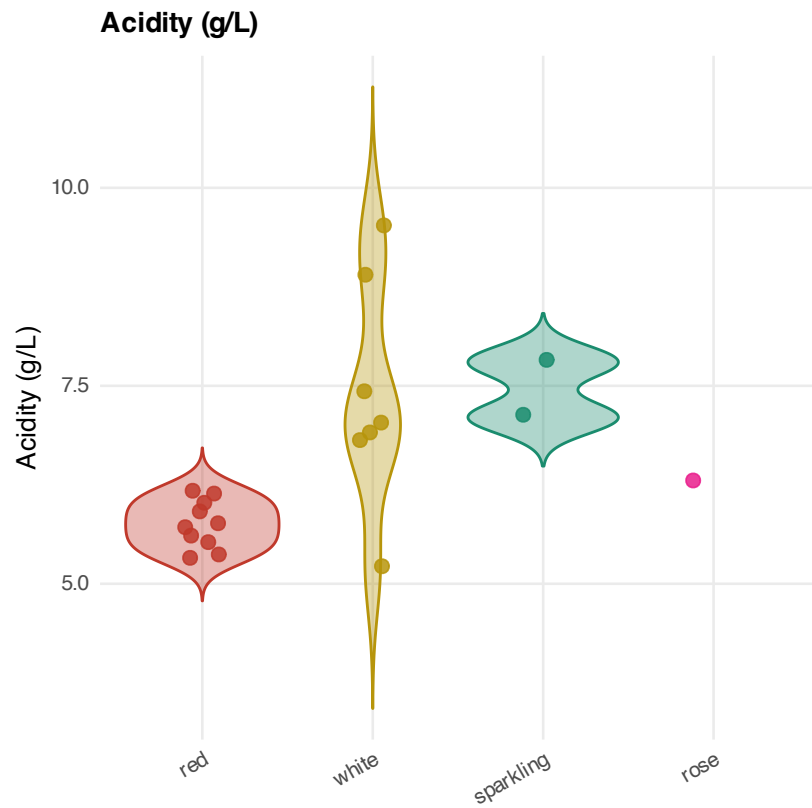
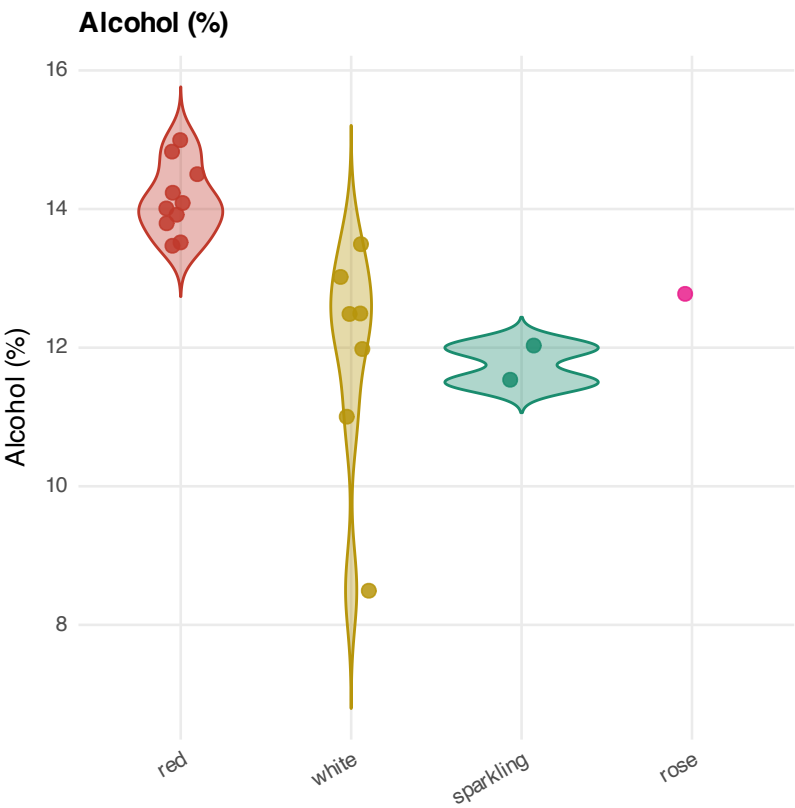


Section 1 — Univariate Distributions

Histograms — Numeric Variables (colour = wine type)



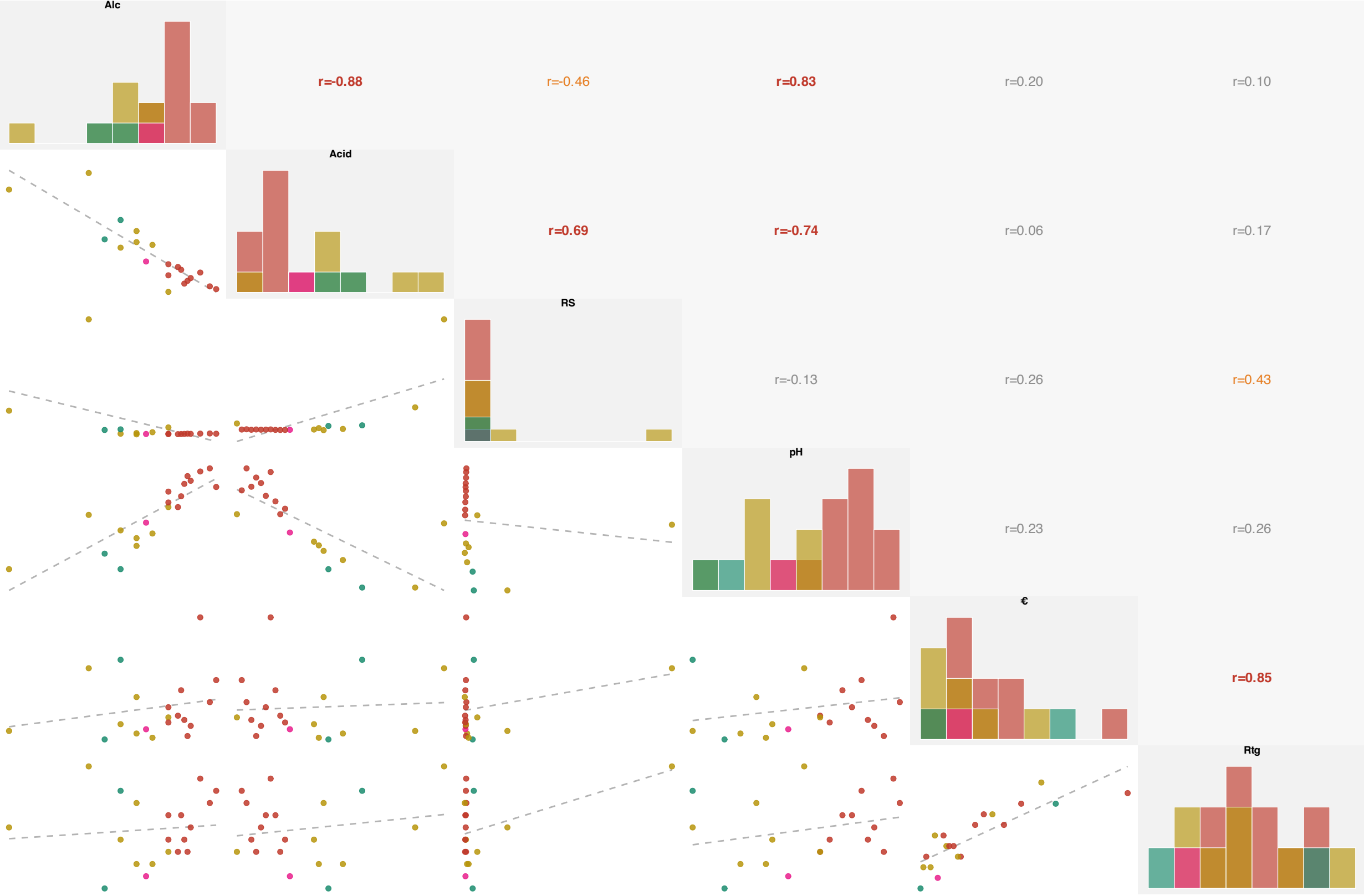
Distributions by Wine Type



Section 2 — Bivariate Relationships

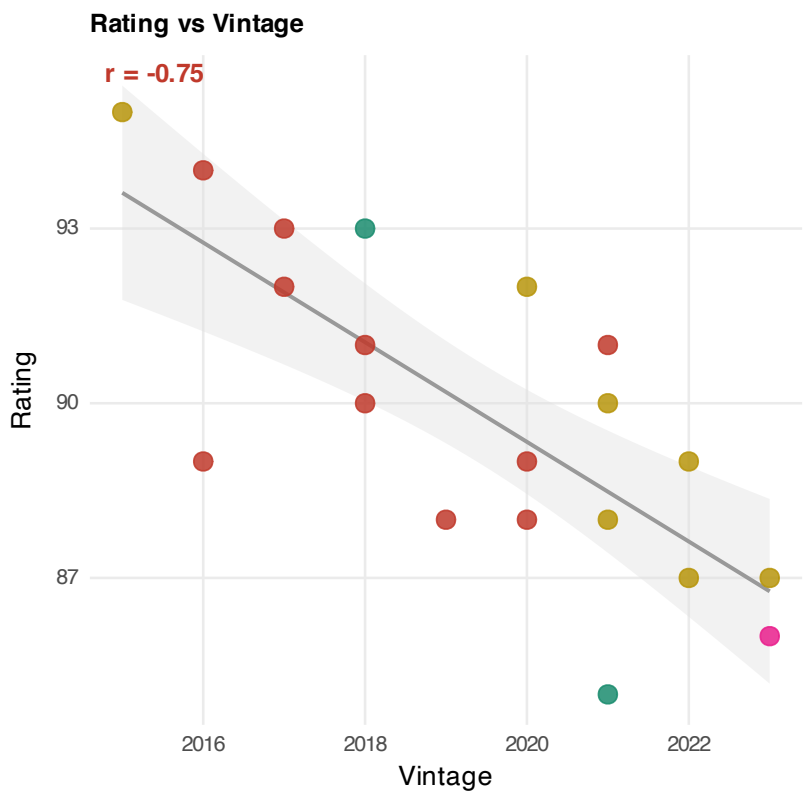
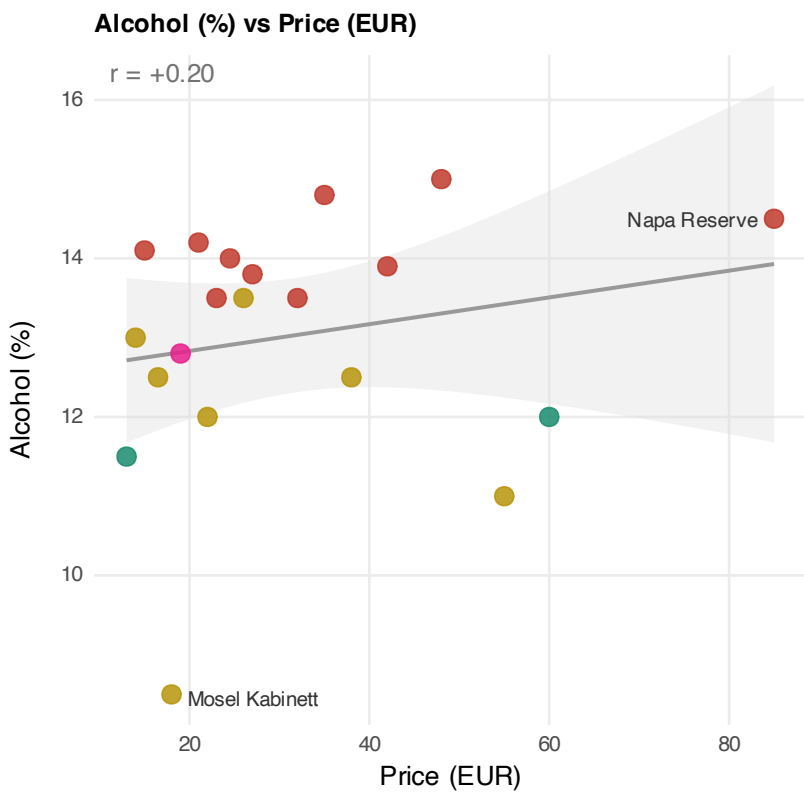
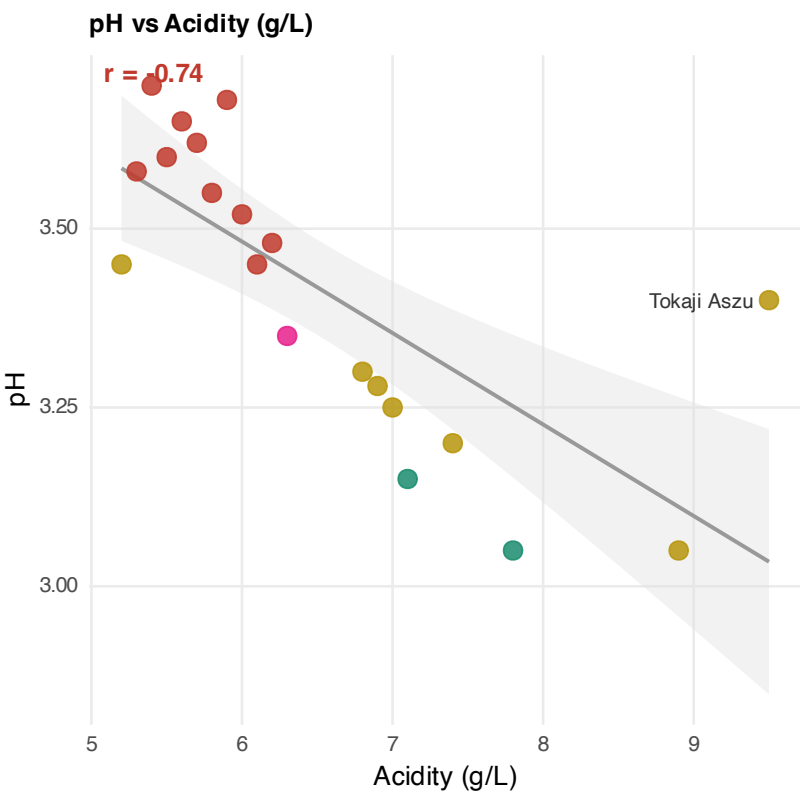
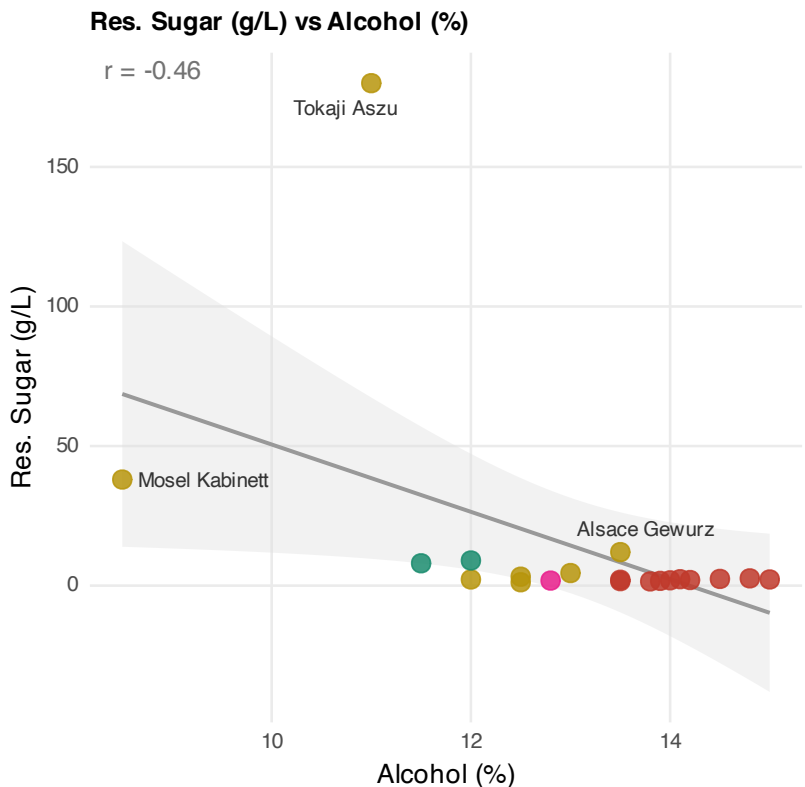
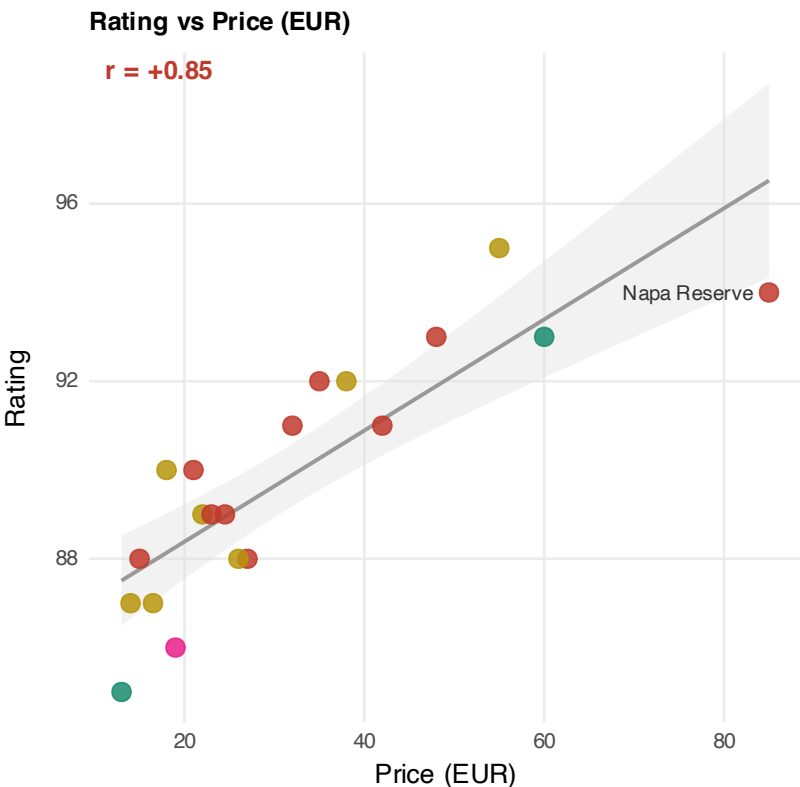
Scatter Matrix (diagonal = histogram, lower = scatter+lm, upper = r)

Red bold $r \geq 0.6$ | Orange $r \geq 0.3$ | colour = wine type



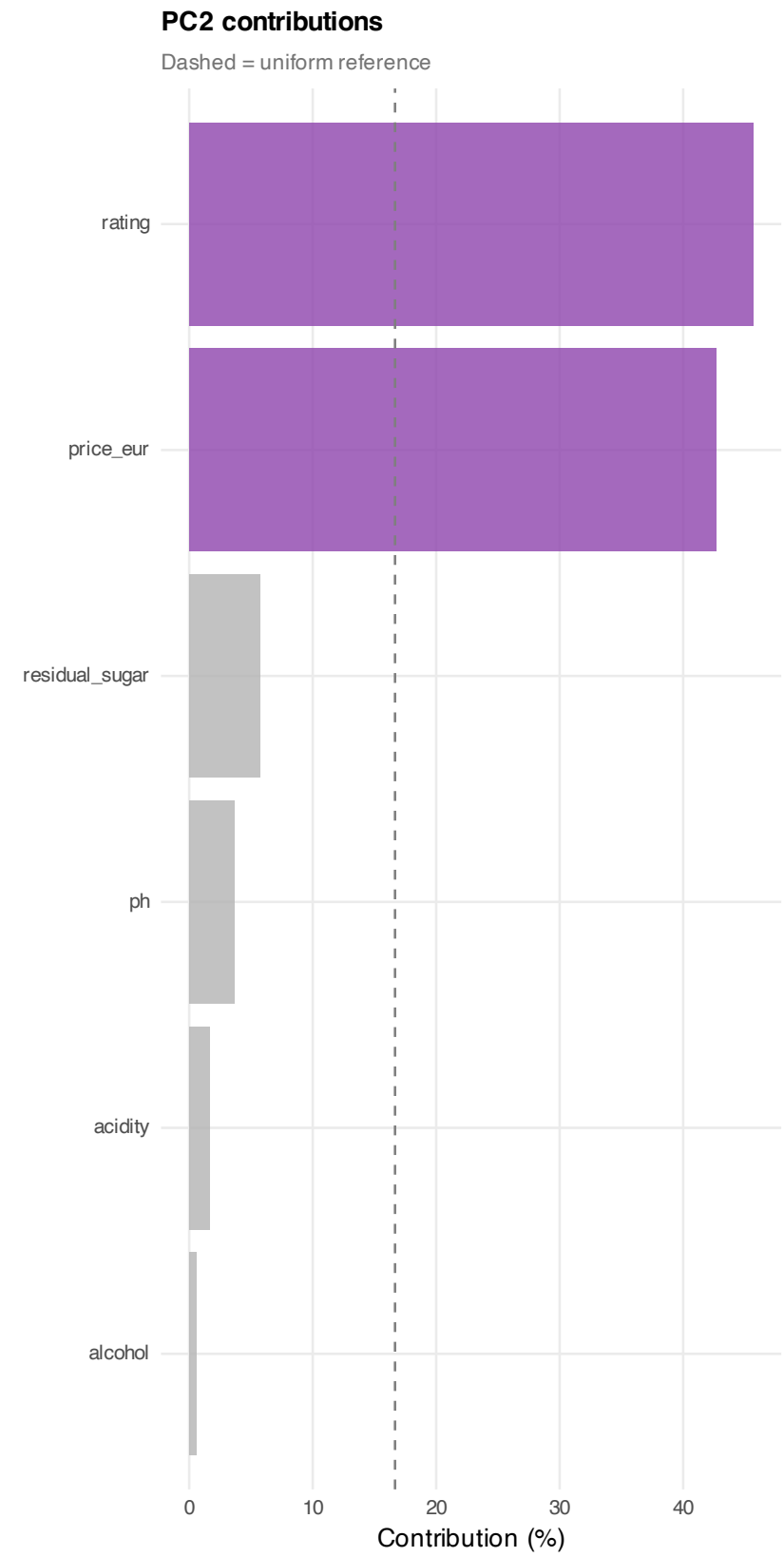
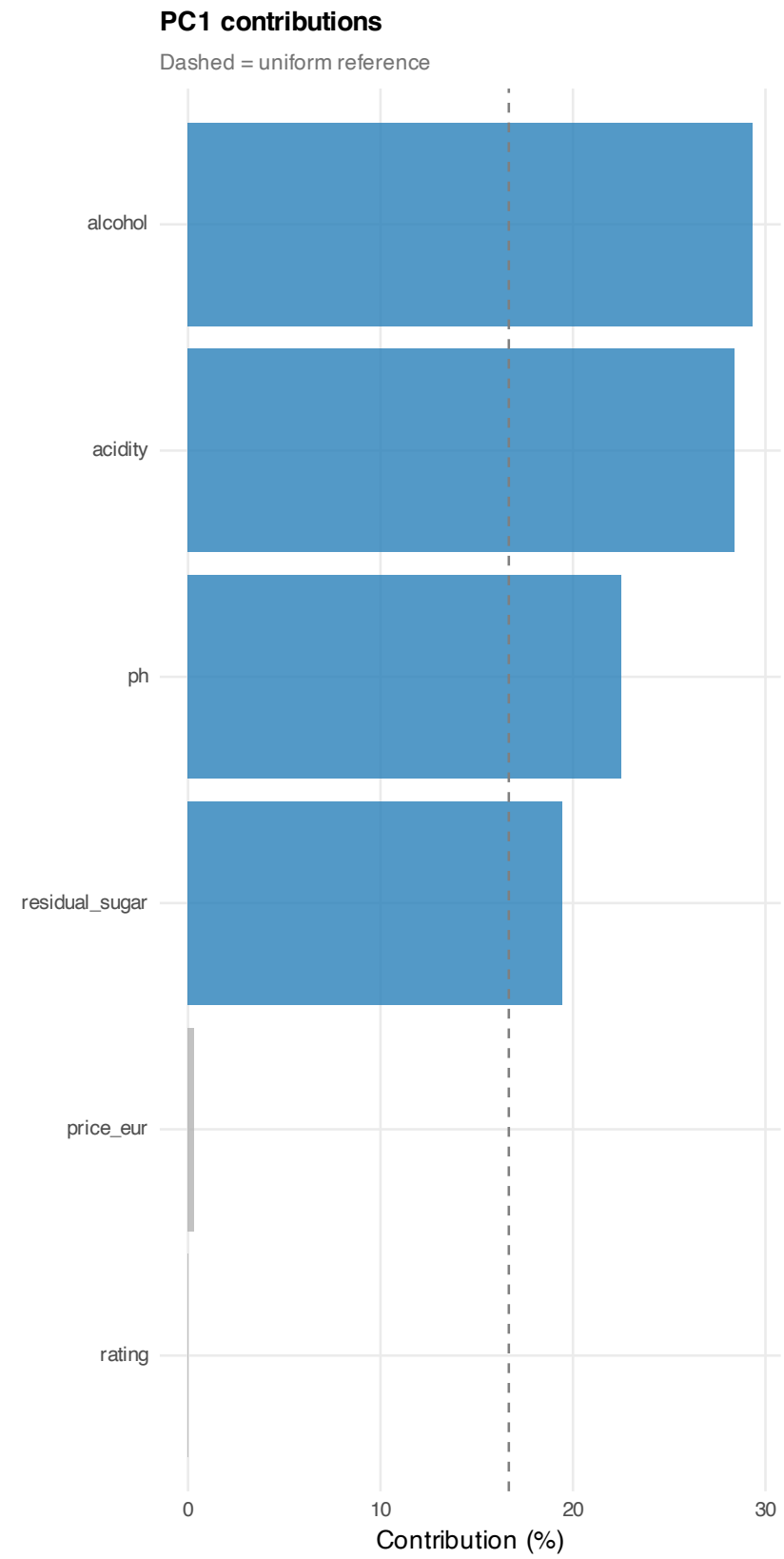
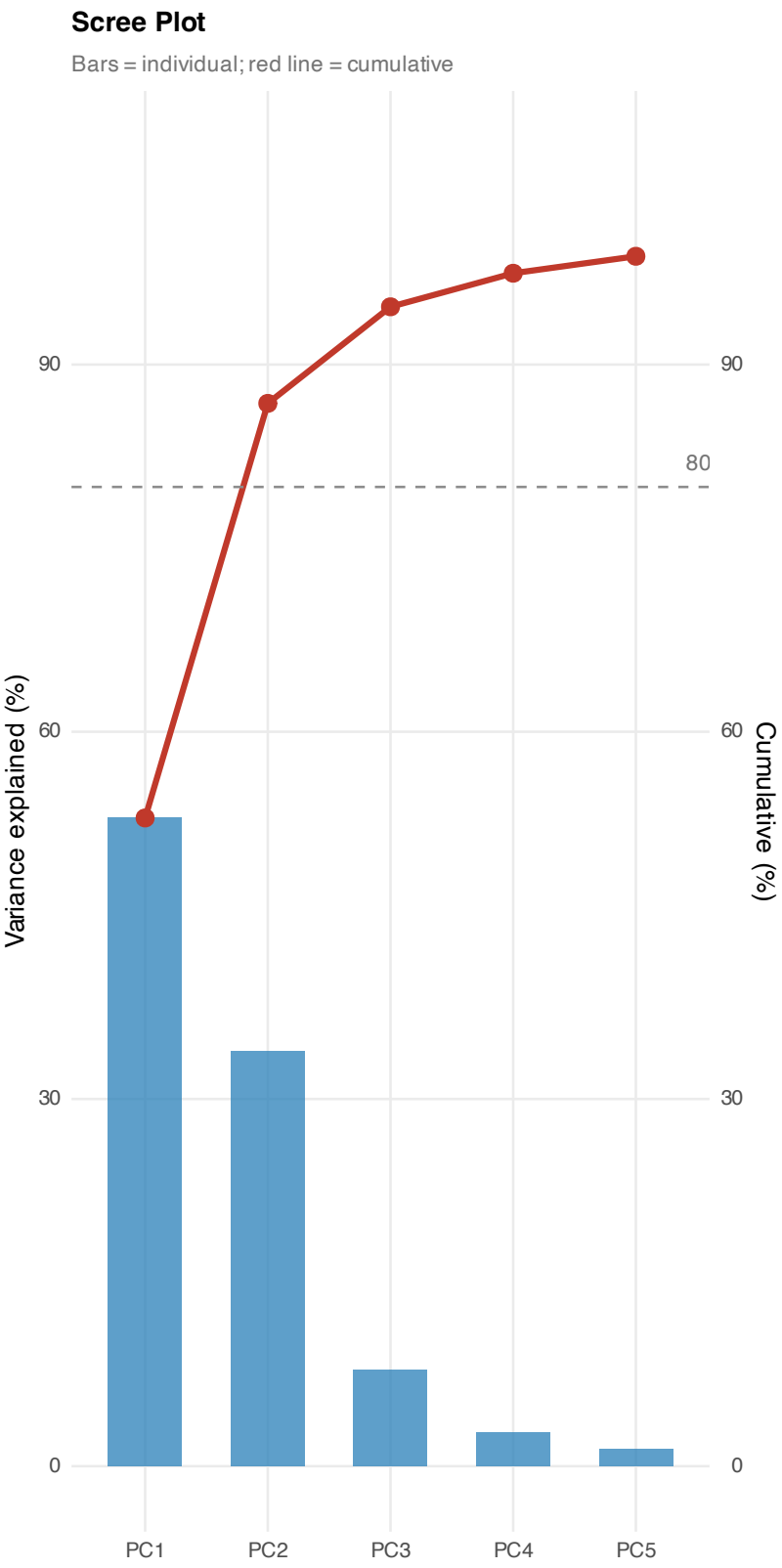
Key Variable Relationships (IQR outliers labelled)

Shaded band = 95% CI of linear fit



Section 3 — Principal Component Analysis

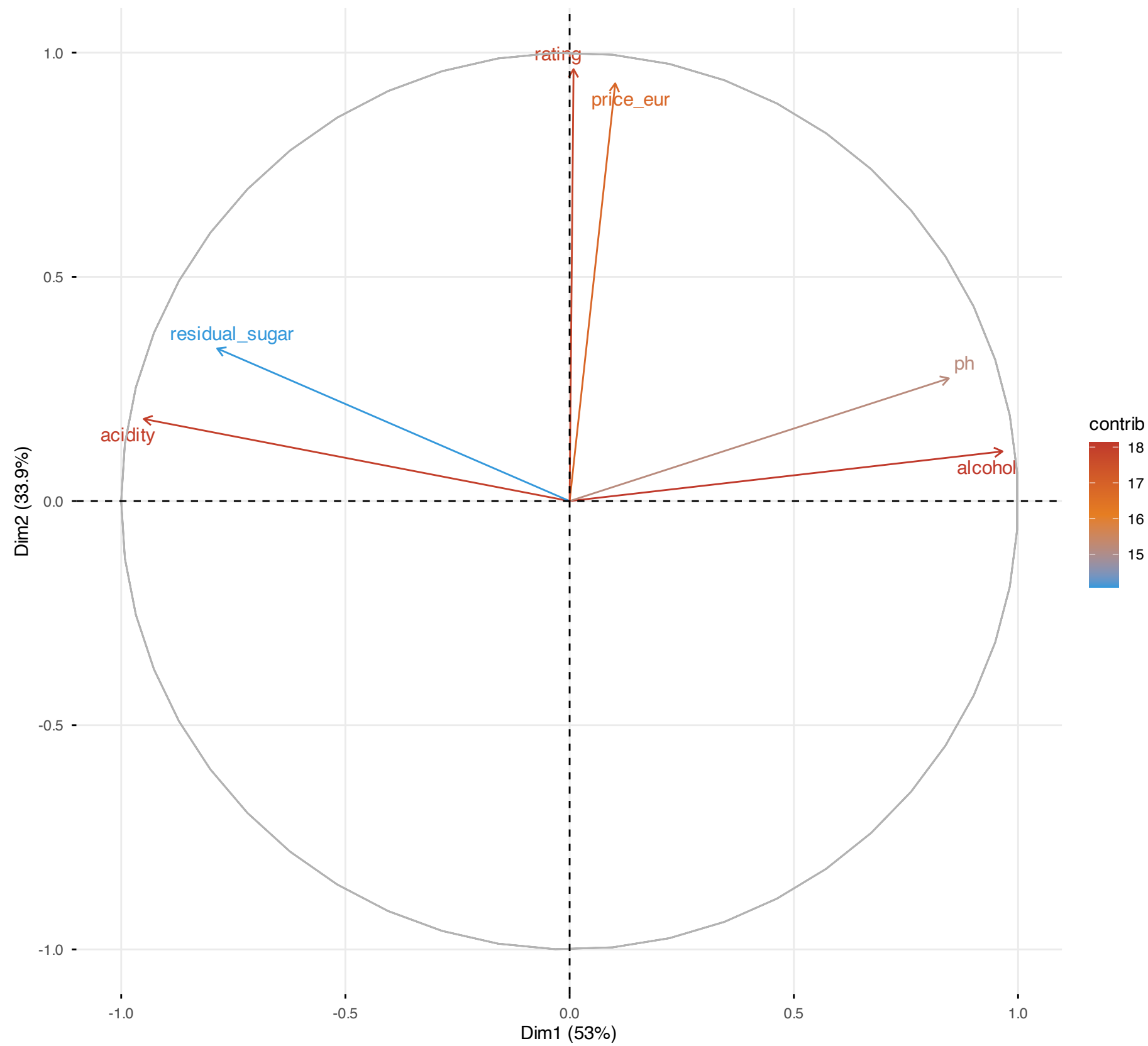
PCA — Variance Explained & Variable Contributions



PCA Biplot — PC1 vs PC2 (residual_sugar log-transformed)

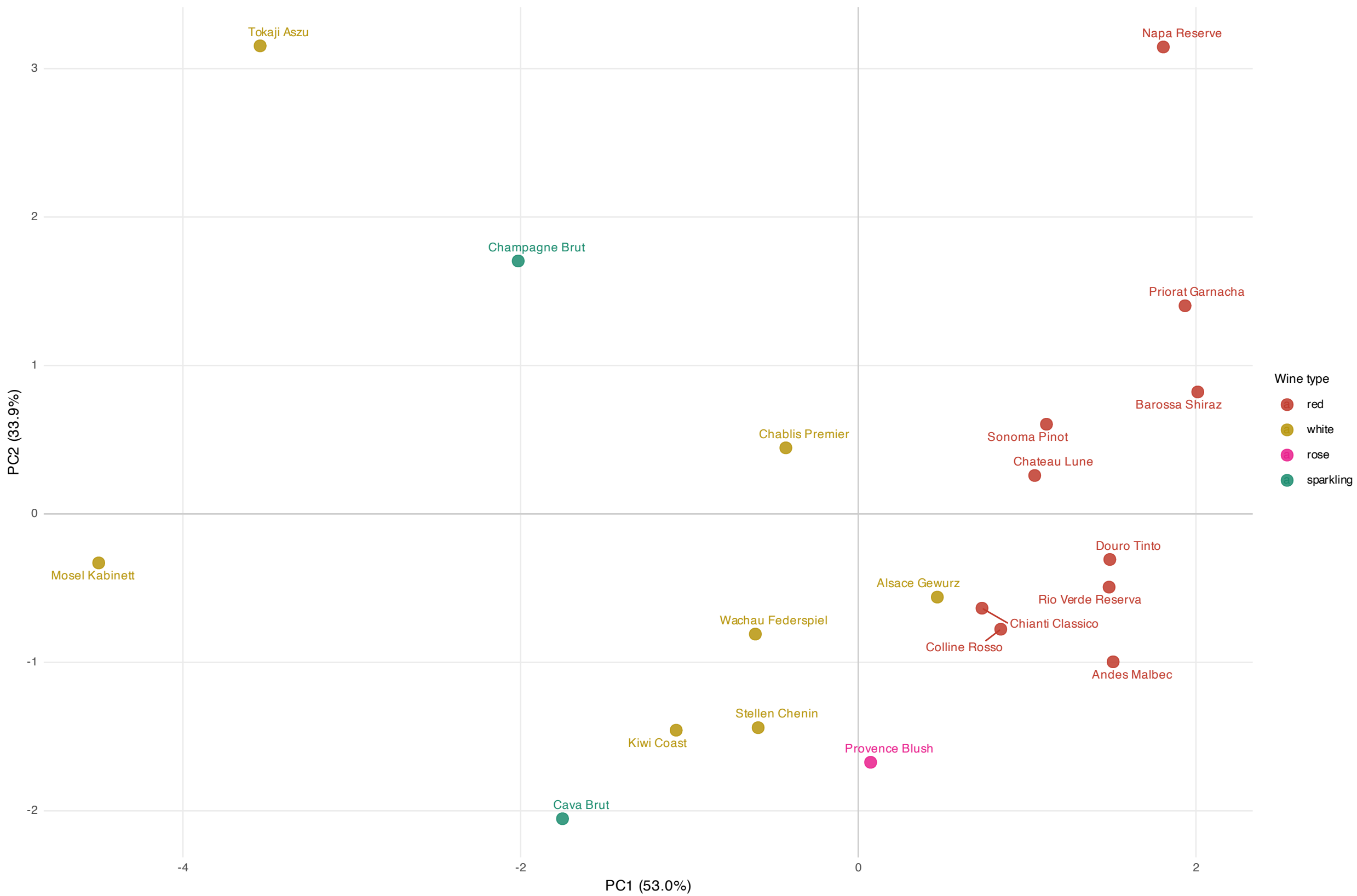


Variable Correlation Circle (colour = contribution %)



PCA — Individual Map

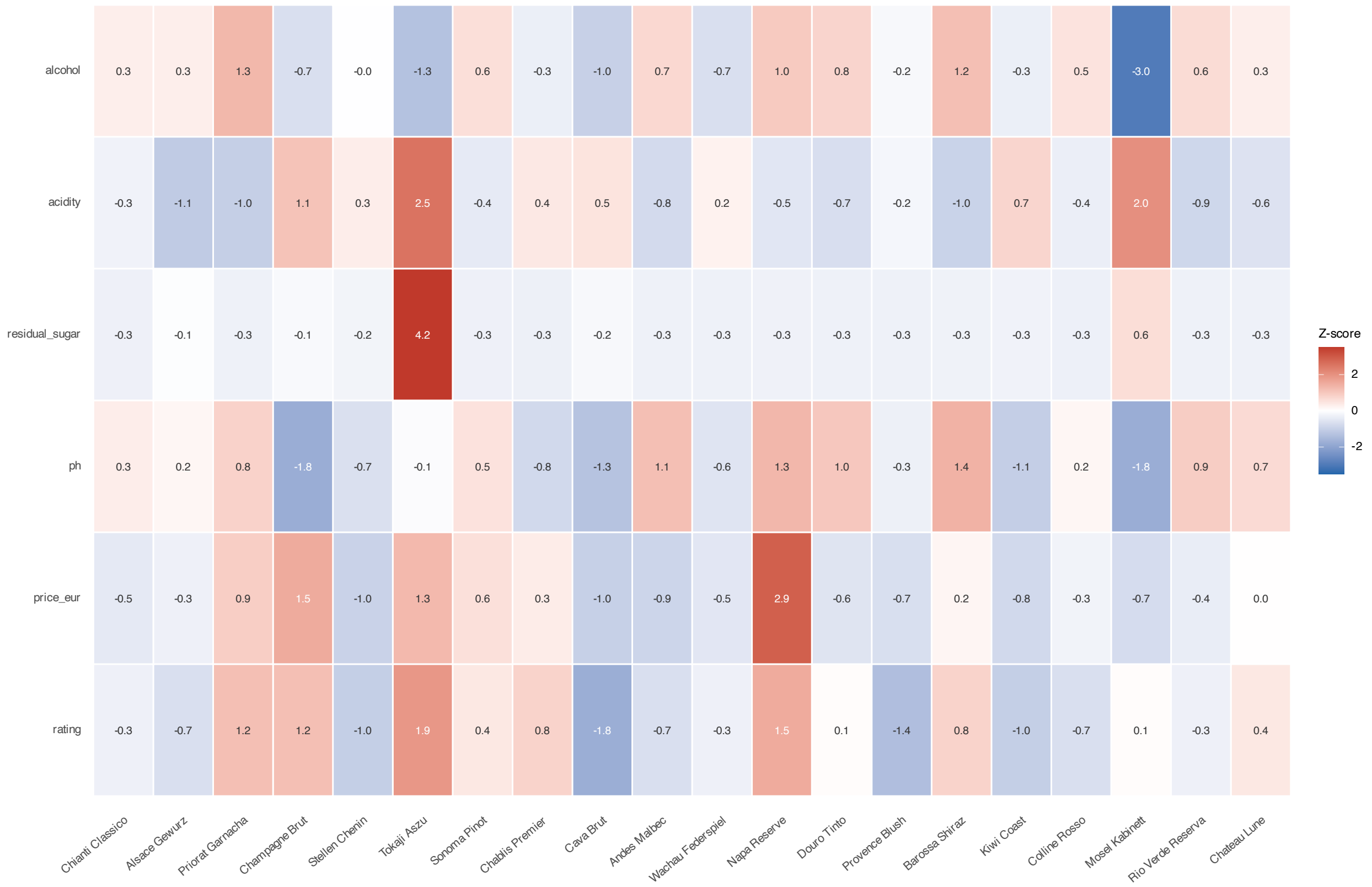
Each point = one wine; coloured by type



Section 4 – Outlier Detection

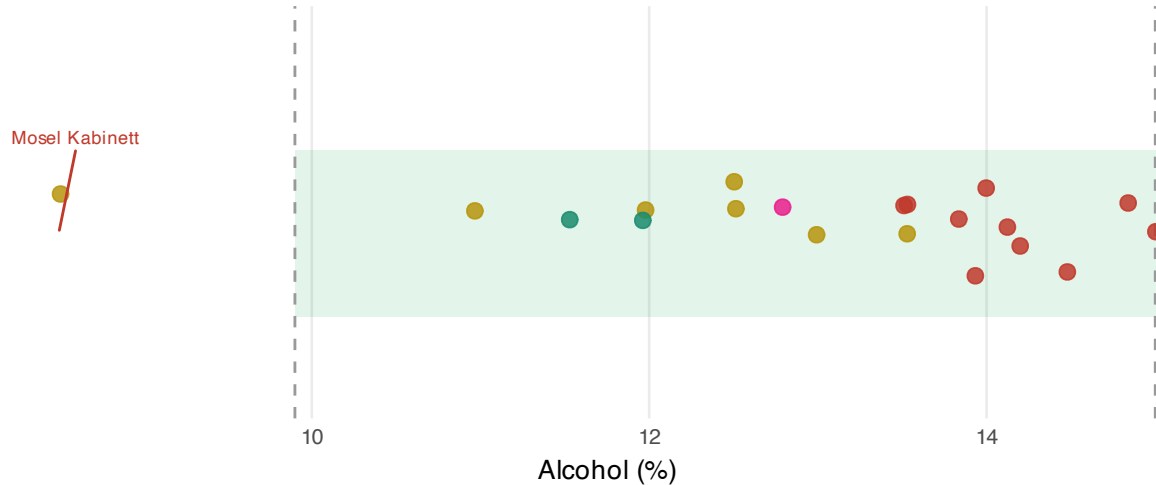
Z-score Heatmap — All Numeric Variables

|z| > 1.5 highlights potential outliers; values capped at ±3.5

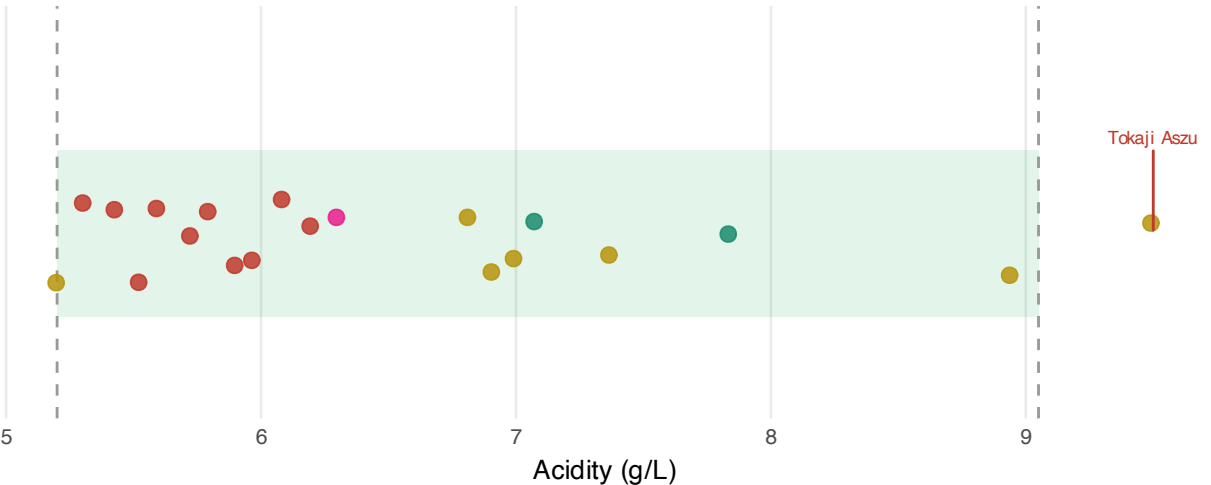


IQR Strip Charts (green band = IQR x1.5 fence; red labels = outliers)

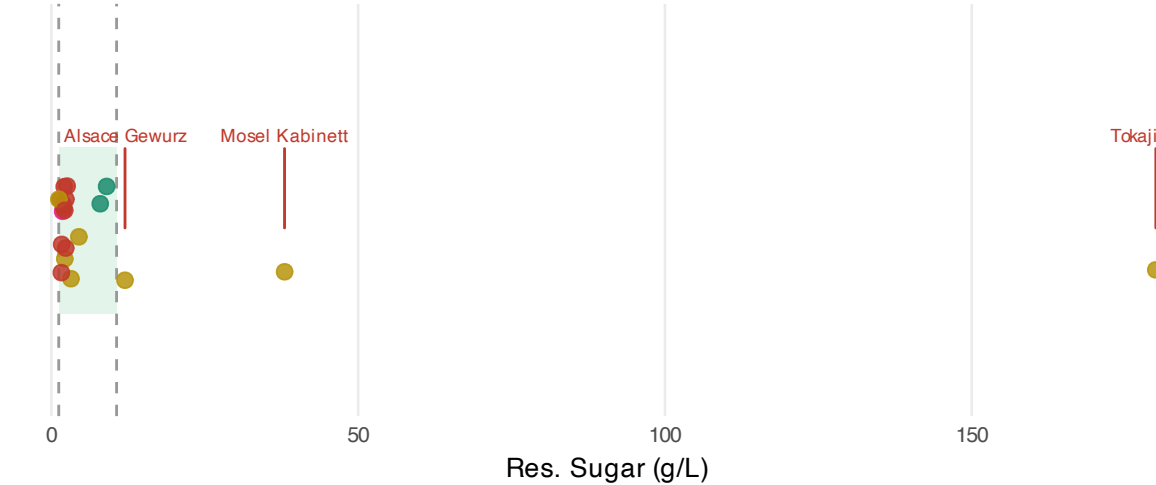
Alcohol (%)



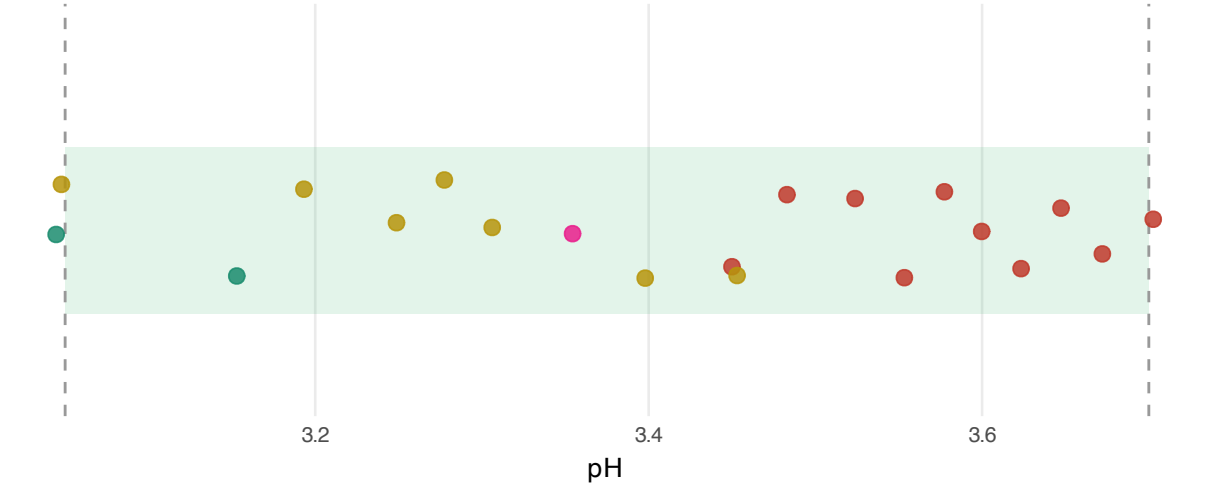
Acidity (g/L)



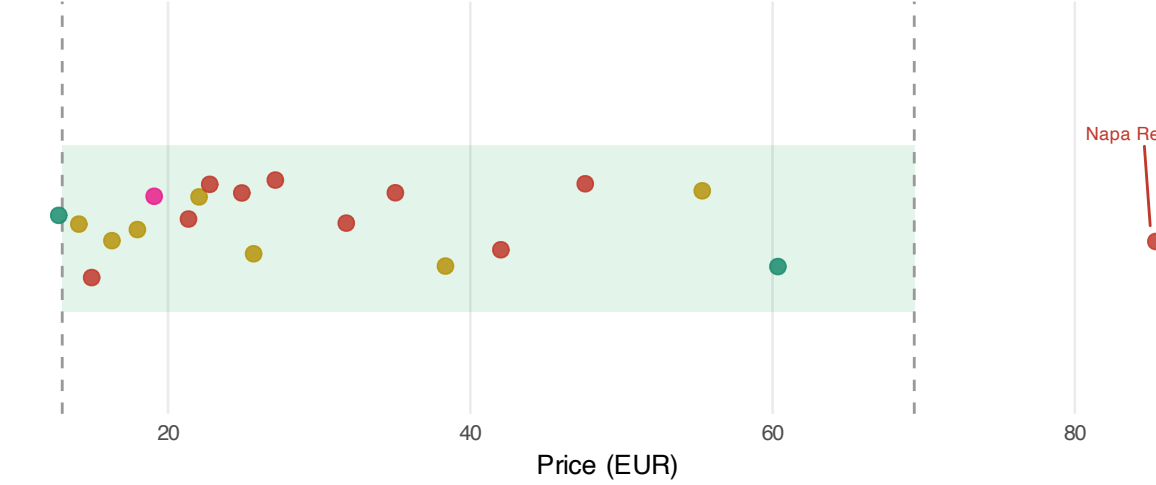
Res. Sugar (g/L)



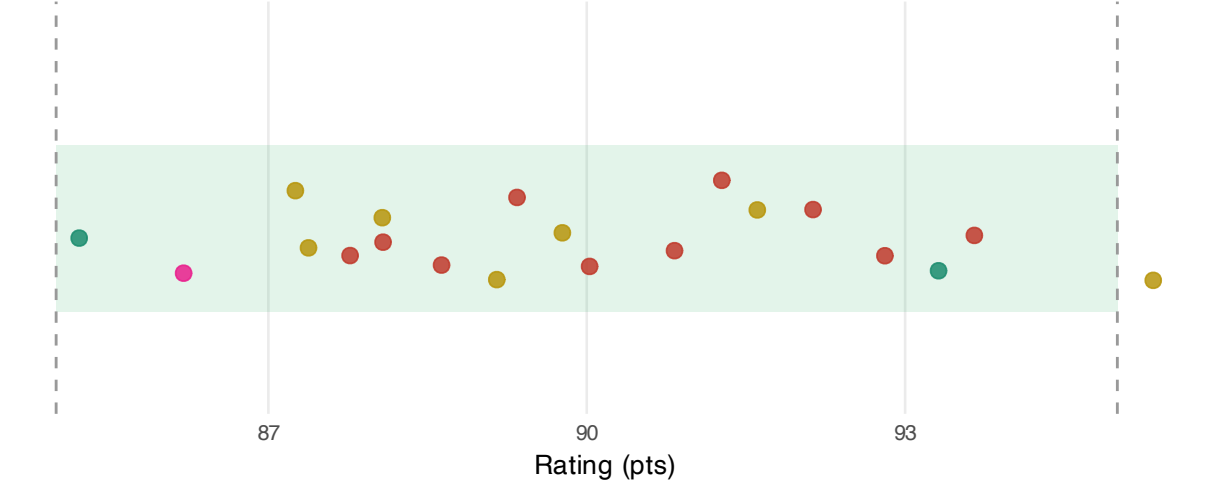
pH



Price (EUR)



Rating (pts)



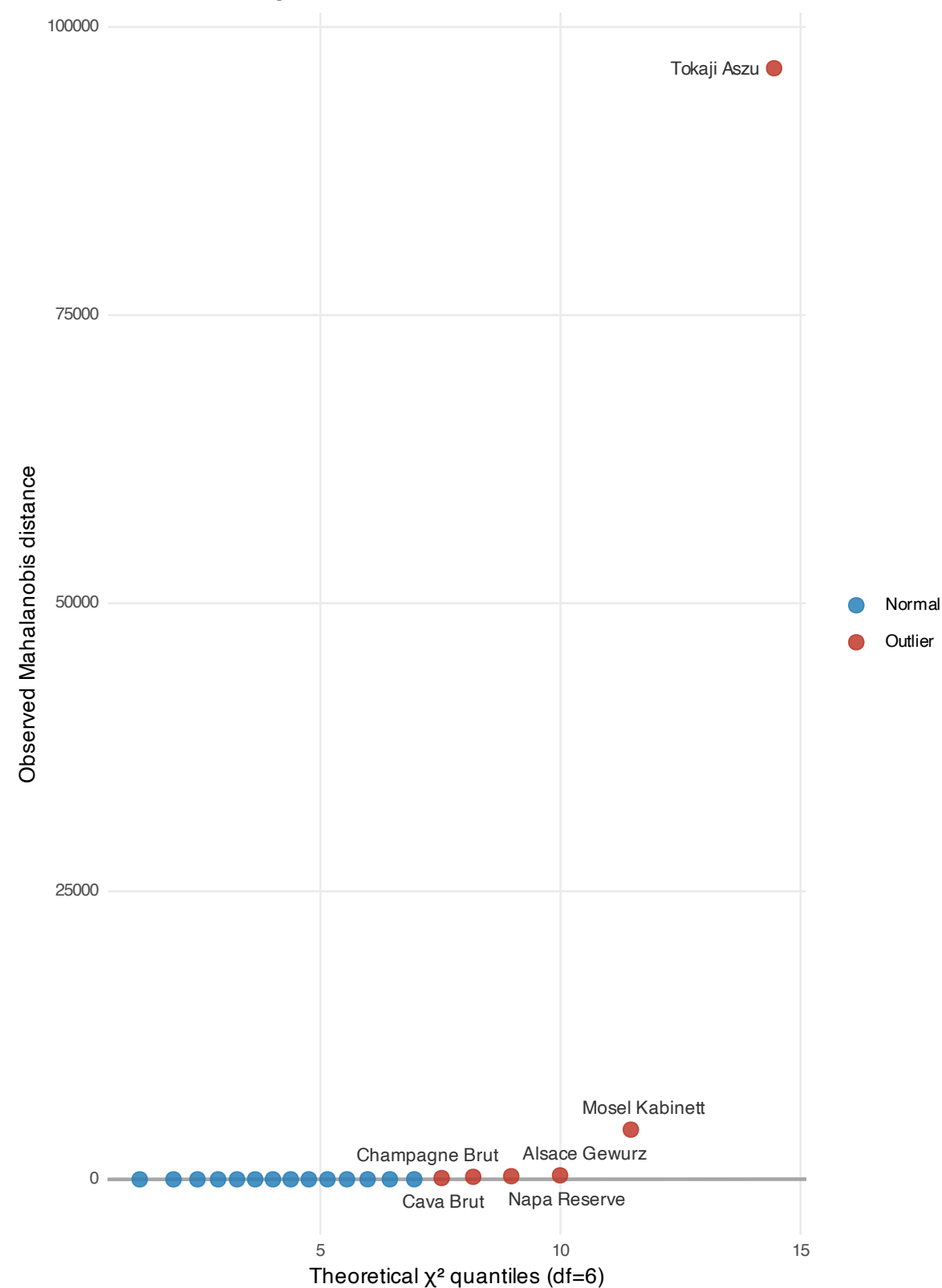
Multivariate Outlier Detection — Mahalanobis Distance

Fully opaque bars exceed the 97.5% chi-squared threshold



Chi-squared Q–Q Plot

Points above the diagonal = heavier tails than multivariate normal



PCA Space — Point Size = Mahalanobis Distance

Bold labels = multivariate outliers beyond $\chi^2_{0.975}$ threshold

